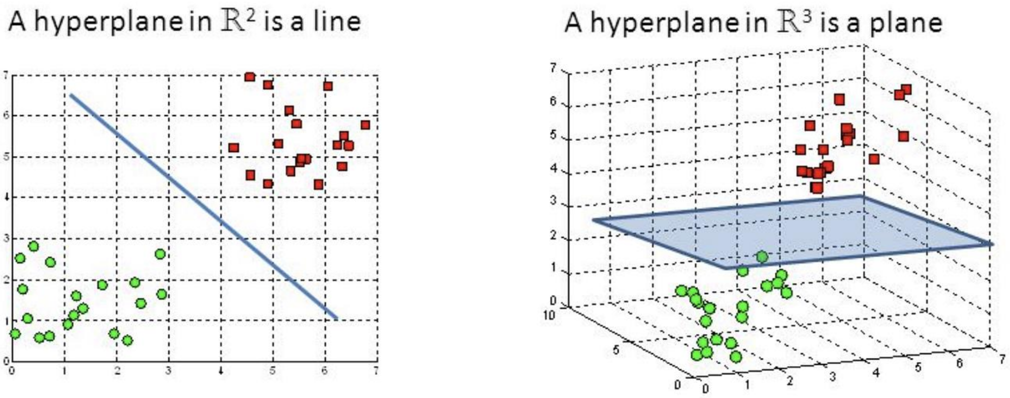

MACHINE LEARNING ASSIGNMENT

IFS 316

ABSTRACT
In-depth analysis on the process of deciding and implementing machine learning concept to a scenario

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Contents

The Scenario Summarized

There is a girl, who will be referred to as the customer, that is trying to find a love interest. She uses online dating as the medium to find her target but faces a big issue. She likes partners in different doses namely in small, large or none. She is not sure how the algorithms pick these options. The filters decided to help determine this is frequent flyer miles, the percentage time spent gaming and the litres of ice cream taken a week.

There many patterns hidden within the data and this will be discussed further within this paper. The topic of Machine Learning which is a subset of Artificial Intelligence which involves the development of machines that can learn by itself backed by systems with mathematical backing. It will be very helpful in identifying these patterns with relative ease in comparison to just doing it by yourself. Clear patterns will be explained and some of the logic behind it will be expanded upon.

Python will be the language libraries due to its extensive libraries that can be used and in the case of this paper for machine learning.

Unprocessed Data

Information is processed data and Machine Learning works best when provided with information. The data that has been provided by the customer shows one clear problem. The filter frequent flyer miles have more of an effect than the others. It is showing greater level of importance compared to the others. Problematic since there was not any requirement to have any of the factors have varying importance so they must therefore have equal level of importance.

The data must be standardised is this will have to be done by normalization, the process of getting data organized to improve the consistency and reliability of them.

Miles has values ranging in the tens of thousands. Percentage time gaming ranges from 0 to lower double digits and the litres of ice cream taken a week ranging in the single digits. A machine learning approaches with their mathematical backing will show results favouring frequent flyer miles over the other features. This will in turn show inaccurate results and relationships and if conclusions are taken from this it will make the customer even more confuse.

To fix this anomaly the normalization technique of minimum-maximum scaling will be used where the maximum value will be scaled to 1 and the smallest value to 0. Python will be used to extract the original file and normalize it. The IDE used will be Pycharm. The mathematical formula for it is $x = \frac{x - x\ minimum}{x\ maximum - x\ minimum}$.

TERMINAL

pip install scikit-learn pandas

pip install xlrd

EXAMPLE CODE

1. Load the original excel file into Pycharm

#1 Replace the string inside the quotes with the exact path from your screenshot
file_path = r"C:\Users\tinob\Downloads\Assignment\exercise2\dating.xls"
df = pd.read_excel(file_path)

 # 2. Create the variable for normalization
scaler = MinMaxScaler()

 # 3. Pick the columns that will be normalized

cols_to_scale = ['MILES', '% TIME GAMING', 'L OF ICE CREAM/WEEK']

 # 4. Duplicate the folder and apply the normalization

df_normalized = df.copy()
df_normalized[cols_to_scale] = scaler.fit_transform(df[cols_to_scale])

 # 5. Save result to a new file
output_path = r"C:\Users\tinob\Downloads\Assignment\normalized_data.csv"
df_normalized.to_csv(output_path, index=False)
df_normalized.to_csv(output_path, index=False)

SAMPLE

MILES	% TIME GA	L OF ICE CI	COMPATIBILITY
0.528341	0.486926	0.564375	largeDoses
0.187063	0.418304	0.99268	smallDoses
0.336372	0.084314	0.475837	didntLike
0.970123	0.768804	0.252057	didntLike
0.495081	0.097642	0.076757	didntLike
0.942453	0.593046	0.611375	didntLike
0.464145	0.399436	0.718599	largeDoses
0.550884	0.776346	0.320421	largeDoses
0.871491	0.504738	0.442614	didntLike
0.458141	0.717683	0.894014	largeDoses
0.648702	0.217734	0.491776	didntLike
0.816979	0.490371	0.990051	didntLike
0.071904	0.285095	0.430347	smallDoses
0.659161	0.273672	0.368813	didntLike
0.998993	0.894654	0.193986	didntLike
0.563886	0.110488	0.110659	didntLike
0.792305	0.439548	0.751897	didntLike
0.899587	0.832648	0.152332	didntLike
0.202311	0	0.740606	smallDoses
0.367824	0.615665	0.773123	largeDoses
0.083757	0.20702	0.486164	smallDoses
0.486869	0.174933	0.492968	didntLike
0.292059	0.309797	0.376596	smallDoses
0.37162	0.385578	0.108175	largeDoses

If certain data points missing?

Future proofing can be needed when certain components of data is missing. Sometimes categorical or numerical data can be missing but a desired result is still being requested from the customer. In this case a Machine learning paradigm of Self-supervised learning will work.

Self-supervised learning is the training of a model to generate future data that is missing based on the already present labelled data. It is a very helpful in predicting data. It combines both supervised learning, where user input is needed to create an output. Aswell as unsupervised learning where data is obtained from unclear data to find patterns that way.

Less human import is needed for the machine to create data. (Bergmann, n.d.) states that although the amount of labelled data already present does affect what kind of output can be delivered. The more labelled data a model must work with then the more accurate predictions it can make.

Some of the benefits are that the model's performance would generally increase, big data can now be more scalable, and the decreased reliance on labelled data.

1. INTRODUCTION: The Machine Learning Approach

We are trying to evaluate data that is historical and have system that is automated with the ability to filter future customer dating options into distinct categories. Namely the people she doesn't like, the ones she likes in small doses and the ones she likes in large doses.

Since the column within the dataset called Compatibility have finalised answers that can be used as input the correct paradigm would be supervised learning. (GeeksforGeeks, 2025) states that supervised learning needs input variables. There is no missing data in the provided dataset and clear labels for categorical and numerical. Essentially an input and output system.

The problem essentially is a Multiclass classification one and there is a single machine learning approach that works well with this. Support Vector Machines, or alternatively SVM, is mathematically backed algorithm that finds patterns in data that has many different dimensions involved in it. Examining human being behaviour in terms of dating is a perfect use for this approach.

2. How SVM functions?

Support Vector Machines uses algorithms that draws an optimal boundary line based on data from the past. It is plotted on a cartesian plane space and the optimal boundary between the different datapoint categories.

Hyperplane and Margin

Hyperplane is a line drawn between different points. A support vector machine doesn't draw any line however cause if it did there would be many lines depending on the amount of data points. The system draws the line with widest possible cap that is called a margin between the different data categories. (Singh, 2024) states there are specific formulas needed that this model use. The model uses the following formula: $wx + b = 0$.

Where w: vector weight (determines what angle, a boundary will be at)

X: data point (the data from the normalized excel document)

B: intercept (how much from graph origin is the boundary shifted)

The model equates the formulae to -1 and 1 respectively to create the margin along with many other calculations. Knowing what formulae, the machines use always helps in understanding how a model gets to a certain point.

This is useful as this new space that is empty between each category can be maximized. When this is done the algorithm makes a new profile that can be classified into the correct group. In the case of dating, you would now know what characteristics the customer prefers.

The Support Vectors

Support Vector Machines get its name from support vectors, which are data points in a cartesian plane that sit on the edge of the margin. They are usually the most difficult to tackle as they are the points that have the smallest distance between other data points from other categories. Support vectors are so important that even if all the other normalized points were removed and only the vectors themselves are kept the exact same boundary will be calculated by the support vector machine.

The Kernel Trick

Human preferences which are usually complex usually needs to be separated by margins that are not straight. To solve this the Radial Basis Function is utilized. Simply it is when instead of just drawing flat lines it is projected into a more complex dimension that is mathematically handled by the computer. This is where the complex behaviours are separated and then it is brought back down to the cartesian plane. The result being a support vector machine that can draw curved boundaries that separate data clusters no matter how inconvenient shaped they are. This makes handling dating preferences with all the 3 distinct categories with ease.

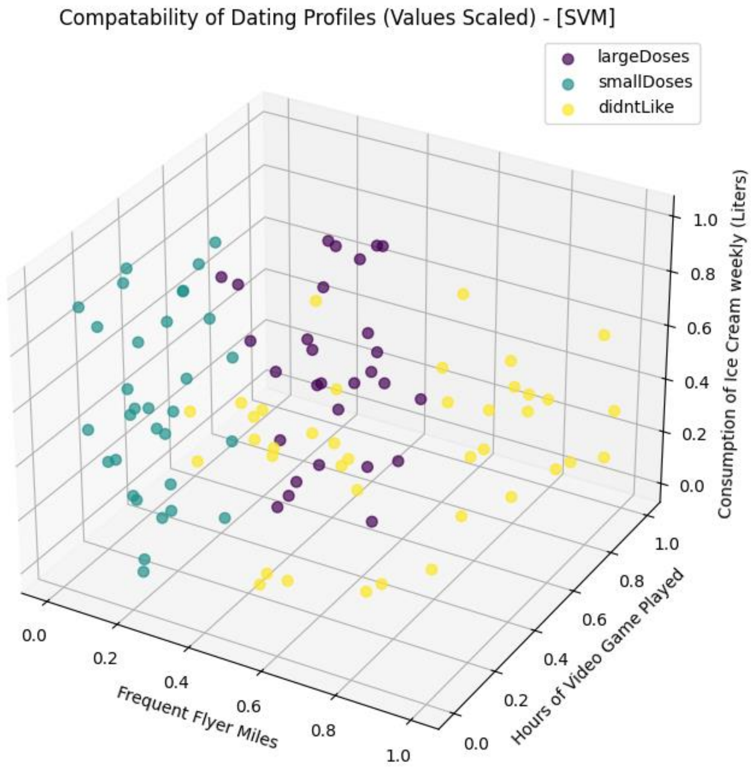
3. Dimensions

How the final visual representation is shown is determined by the number of variables that are being analysed. For a two-dimensional space there obviously would be a flat diagram but in the case of the customer there are three dimensions, namely the consumption amount, frequent flyer miles and time spent on videogames.

The True 3D Scatter Plot

The true representation of the dataset given is in a three-dimensional space or scatter plot. It is the only accurate way to represent the dating profiles at the same time and identify the optimal boundary between them. The optimal feature boundary will have a X, Y and Z axis.

This is how the support vector machine algorithm visualized the data. You can see the literal data points clustered in a three-dimensional world. There are two-dimensional hyperplanes between data categories that separate the clouds of points.



Pycharm generated three-dimensional plot of an SVM

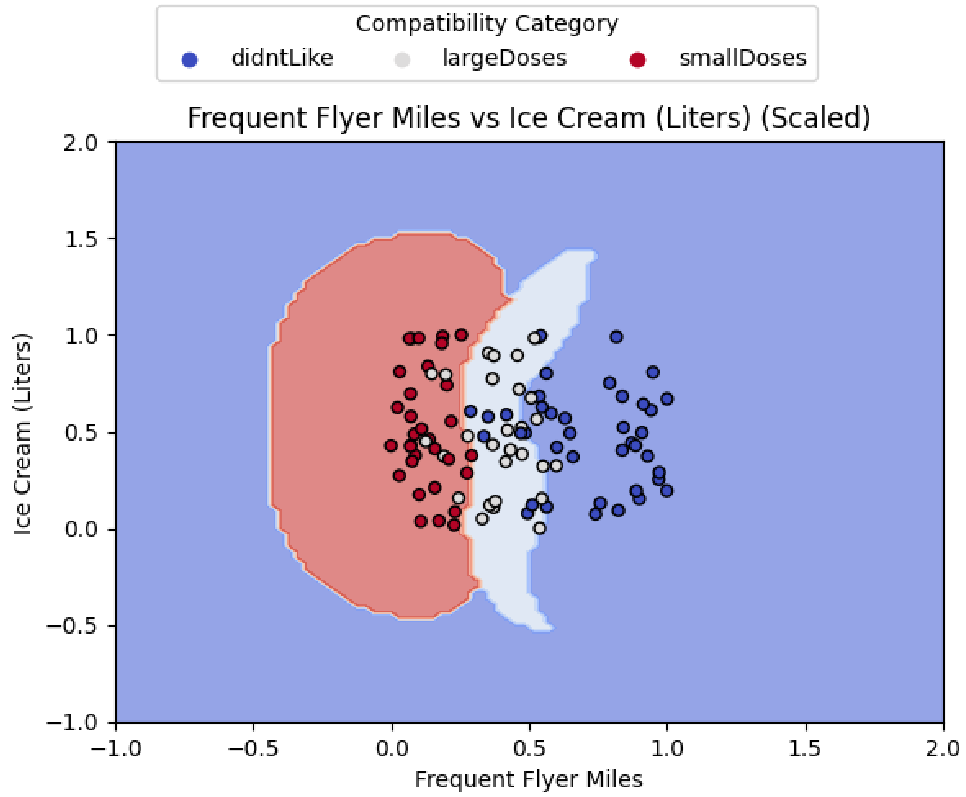
Use of 2D Scatter Plots

Three-dimensional scatter plots are the mathematically correct and it is very important to understand how a machine sees your data. But for easy human interpretation 2d scatter plots are very useful. It is difficult at times to see the behavioural patterns between clouds and clouds of data points. Generating two-dimensional pair-plot matrices would help immensely.

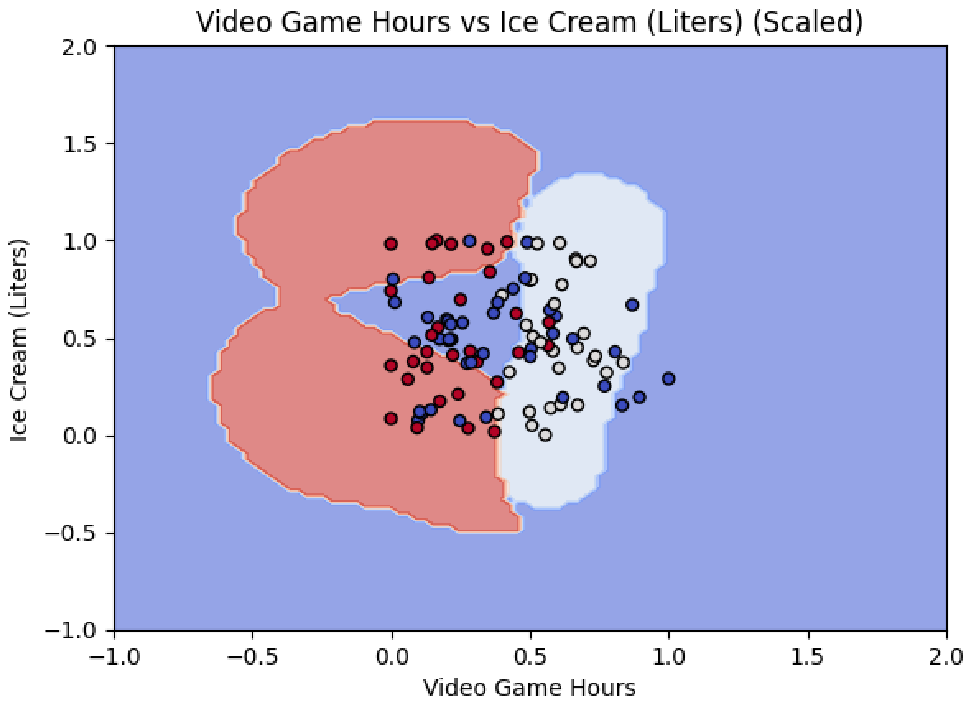
Since we have 3 data categories namely, ice cream consumed, frequent flyer miles and time spent gaming. Three separate graphs comparing two features at a time would help identify relationships even better. For example, time spent gaming versus ice cream consumption in litres.

We can in a two-dimensional space colour in territories as the algorithm will now be asked to display the data in two-dimensions as well. The common human limitation of three-dimensions interpretation of a scatter plot can be removed entirely. Simply if data point falls in the red zone, then it is not a good match for the customer, if it falls in green zone, it is a near perfect match. These colour separations are vital as identifying hidden patterns become so much easier this way then ever before. The customers hidden preferences can be identified.

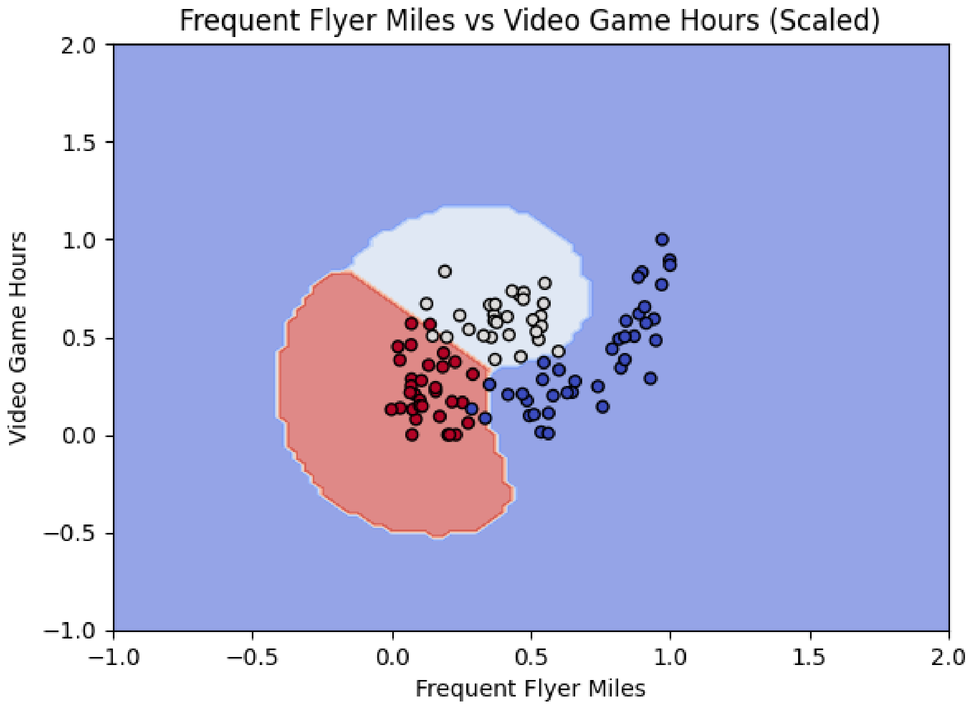
(GeeksforGeeks, 2025)



Ice Cream Consumed versus Frequent Flyer Miles



Ice Cream Consumed versus Video Games Played



Video Games Played versus Frequent Flyer Miles

4. Determine Compatibilities

Both the two-dimensional and three-dimensional graphs will be utilized to determine the optimal boundaries within them. The exact dating preferences of the customer can be identified.

The didn't like filter

The support vector machines algorithm clearly sectioned of what would be classified as an automatic rejection.

Extreme gaming habits was one that was an automatic rejection. When video game hours are over a certain amount of time it becomes an automatic rejection for the customer. Lack of travelling was another automatic rejection. When frequent flyer miles were compared to other filters it was seen that a low amount was mostly covered by the didn't like territory. This shows that no matter the amount of ice cream consumed or gaming habits if they rarely travel the customer doesn't deem them worthy.

Small Doses Filter

This is the middle ground in the visualizations. A certain balancing act is seen to be needed for small doses. When you look at frequent flyer miles for a small dose, they need to have an amount that is over the automatic rejection phase, but not so high that they are seen as exciting and highly qualified to be with the partner. For hobby type filters like the litres of ice cream consumed and gaming played a balancing act is needed.

Large Doses Filter

This is where the support vector machine found maximum compatibility with the customer. The hidden patterns being found by the algorithm. On the scatter plots specific territories for the ideal matches were carved by the support vector machine.

One filter called frequent flyer miles are shown to have to be extremely high to be compatible. In both the three-dimensional spaces and the two-dimensional spaces, high frequent flyer miles take the upper ranges of the cartesian planes. The customer clearly likes partners that live a dynamic life experiencing new things often. But they want balance as well in life. They need to game as well. Ice Cream was found to be a filter that barely effects the other factors and can be concluded as something that isn't a factor that effects what the customer likes.

5. FINAL INSIGHTS

There is still a balancing act needed for the other filters. Leisure habits like video games must be in the lower bounds. They have such a big effect that even if the frequent flyer miles are high if video game usage is high as well the points won't be as compatible for the customer. The partner essentially can game in their free time but mustn't exclusively do it and must be outgoing.

The results of implementing a support vector machine allowed us to turn what looked like a random excel folder into something that can be interpreted and improve the situation of a customer. The mapping of human behaviours with the kernel system being able to separate categories and we are being able to decipher them. It was found that high video game usage with low frequent flyer miles was an automatic rejection for the customer. This shows that for future customers there is an algorithm that can determine what a person likes in dating prospects.

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